

Robert Smith

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EDUCATION

University of Florida

Bachelor of Science, Computer Science

2025

The University of Texas at Austin

Master of Science, Computer Science

Expected 2027

EXPERIENCE

Python Software Engineer

Aug 2025 to Oct 2025

Tort Experts

Remote

- Developed Python software automating customer data ingestion and processing.
- Designed Prefect orchestration pipelines integrating heterogeneous datasets into GCP.
- Designed relational schemas supporting scalable production analytics.
- Diagnosed and resolved production pipeline issues, improving system reliability.

Technical Consultant

Jul 2023 to Dec 2023

Alpha Omega

Remote

- Documented organizational structure, engineering teams, technology stack, and infrastructure of NOAA NCEI's Data Access Branch as part of a technical assessment.
- Produced technical documentation describing current architecture and future infrastructure modernization objectives for strategic planning.

Data Engineer

Jun 2022 to Jan 2023

Socure

Remote

- Developed Python data services processing hundreds of millions of fraud records.
- Increased ingestion throughput from 20 million to more than 250 million records through pipeline optimization.
- Built Airflow workflows integrating AWS services and Snowflake.
- Developed CI/CD pipelines and optimized SQL performance, reducing dashboard latency by 40 percent.

Full Stack Developer

Feb 2020 to Sep 2021

Riverside Technologies / NOAA NCEI

Remote

- Developed production software using Python, React, Node.js, SQLAlchemy, and relational databases.
- Designed REST APIs and normalized database schemas supporting scientific data systems.
- Collaborated with stakeholders to translate technical and functional requirements into production software.

PROJECTS

STM32F407 Quadcopter Flight Controller | C, ARM Cortex M4, Bare Metal

- Developing a bare metal flight controller for an STM32F407 based quadcopter using register level C without HAL based peripheral drivers.
- Implementing reusable GPIO and I2C drivers from the STM32F407 reference manual and peripheral register specifications.
- Integrating an MPU6050 IMU over I2C for accelerometer and gyroscope acquisition, configuration, and flight state sensing.
- Designing firmware architecture for sensor acquisition, PID based self leveling, receiver input processing, and motor control.

- Using datasheets, reference manuals, and hardware debugging techniques to validate communication between the microcontroller and external devices.

STM32 LoRa Telemetry System | C, Python, FreeRTOS

- Developed embedded firmware integrating MPU6050 sensor acquisition with UART, SPI, and I2C communication.
- Implemented STM32 peripheral drivers and integrated LoRa wireless communication for telemetry transmission.
- Used FreeRTOS tasks and queues to separate sensor acquisition, communications, telemetry, command, and system health functionality.
- Used a logic analyzer to inspect digital signals and validate communication between embedded devices.
- Developed a Python ground station for decoding, visualizing, and logging telemetry data.

Operating System Kernel and Virtual Filesystem | C, x86 Assembly

- Developed a 32 bit operating system kernel in C and x86 assembly with memory management, scheduling, terminal functionality, and filesystem support.
- Designed a virtual filesystem abstraction using function pointers and modular interfaces for multiple filesystem implementations.
- Worked directly with low level memory, hardware interfaces, interrupt handling, and cross compiled system software.

TECHNICAL SKILLS

Languages: C, C++, Python, x86 Assembly, SQL

Embedded: STM32, ARM Cortex M, Bare Metal Firmware, FreeRTOS, GPIO, UART, SPI, I2C, Interrupts, PWM

Hardware: MPU6050 IMU, LoRa, Logic Analyzer, Microcontroller Peripheral Integration

Systems: Linux, RTOS, Embedded Systems, Device Drivers, Multithreaded Programming

Tools: STM32CubeIDE, GCC, GDB, Git, CMake, Make, QEMU, Docker, AWS, GCP